

Enhanced Drone-based Plant Disease Detection with Visual Data Augmentation and Optimized Training

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Abstract—This paper presents an enhanced detection model utilizing data augmentation and learning rate optimization over the YOLO detection framework, applied to drone-captured imagery of cashew orchards. We focused on three stressors: insect damage, biotic stress from pathogens, and abiotic factors. To address dataset imbalance, advanced data augmentation was implemented alongside a dynamic learning rate adjustment for optimal performance. We evaluated two object detection models, YOLOv9 and YOLOv10. YOLOv9 achieved an mAP50 of 0.591 for insect detection, 0.512 for abiotic stress, and 0.512 for disease detection. YOLOv10 improved abiotic detection with an mAP50 of 0.773 but showed lower performance for insect stress (0.464) and disease (0.476). These results demonstrate the improvements achieved through our enhancements, highlighting the need for diverse datasets and better visualization techniques in precision agriculture.

Index Terms—Crop Disease Detection, YOLOv10, YOLOv9, Object Detection

I. INTRODUCTION

The rapid advancement of computing technologies, particularly in machine learning (ML) and deep learning (DL), has transformed agriculture by shifting plant disease detection from labor-intensive methods to automated, data-driven approaches. This shift is driven by the need for more efficient, scalable solutions to maintain global food security and improve crop yields. Traditional plant disease detection methods rely heavily on manual inspections, which are time-consuming and prone to error, making them impractical for large-scale agricultural operations.

Drone technology, combined with deep learning models, has introduced a new era in plant disease detection by enabling large-scale, real-time monitoring of agricultural fields. Drones equipped with high-resolution cameras can capture detailed imagery, allowing for early detection of diseases that might otherwise go unnoticed at ground level [1]. The ability to efficiently monitor crops using drones has become crucial in modern precision agriculture, where timely intervention can prevent disease outbreaks and minimize crop losses [2]. Despite the benefits, challenges remain in adapting computer vision methods to real-world agricultural environments, where

variables such as lighting conditions, crop growth stages, and weather patterns introduce complexity [3], [4].

Current computer vision techniques, particularly deep learning models like YOLO (You Only Look Once), have set a new standard in real-time object detection and are well-suited for agricultural applications [5]. However, applying these models to agriculture presents unique challenges. Existing methods struggle with handling diverse datasets under varying conditions, which leads to difficulties in generalization and accuracy. Moreover, optimizing these models for real-time, on-the-fly disease detection is computationally demanding and requires innovative solutions to overcome performance bottlenecks.

To address these challenges, we propose the use of enhanced YOLO models—YOLOv9 and YOLOv10—specifically optimized for plant disease detection using drone-captured imagery. Our research introduces a comprehensive framework (as shown in Figure 1) that integrates two key technical advancements: advanced data augmentation and automated learning optimization. Data augmentation generates varied and balanced training samples from the cashew dataset, helping address issues such as class imbalance and overfitting [6], [7]. Additionally, we implemented an orchestrated learning rate training function that dynamically adjusts the learning rate, improving model performance in dynamic agricultural environments [8].

By leveraging these innovations, YOLOv9 and YOLOv10 are designed to meet the computational demands of large-scale, multi-class datasets while maintaining high efficiency and accuracy. Our approach aims to push the boundaries of real-time disease detection in precision agriculture, enhancing crop health monitoring and enabling timely interventions. This research supports the development of AI-driven solutions alongside with deep learning solutions that autonomously monitor crops, optimize resource allocation, and provide actionable insights for farmers, ultimately contributing to more sustainable and resilient food systems.

The structure of this paper is as follows: Section 2 provides an overview of related work on plant disease detection using machine learning and drone imagery. Section 3 outlines our methodology, covering dataset preparation, model architecture, and optimization strategies. Section 4 presents the experimental results along with a detailed analysis. Lastly, Section 5

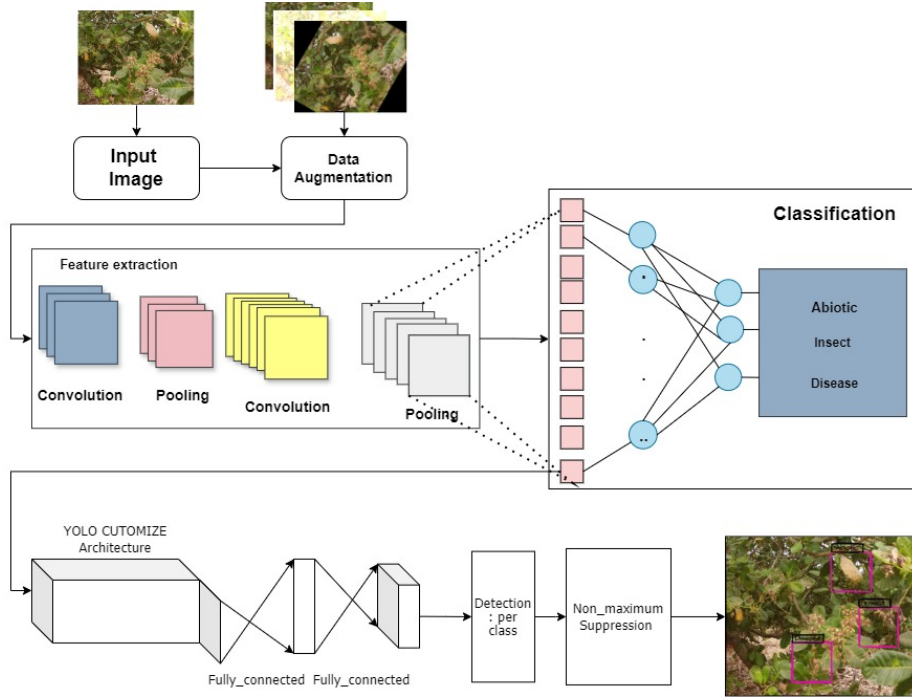


Fig. 1. Proposed Framework for Plant Disease Detection Using YOLO. The framework incorporates data augmentation techniques to enhance model training, followed by an optimized learning process applied over the customized YOLO architecture. This setup is designed to detect and classify stress factors such as abiotic stress, insect damage, and diseases from drone-captured imagery, with non-maximum suppression applied to refine detections.

concludes the paper, discussing key findings, limitations, and directions for future research.

II. RELATED WORK

The detection and prediction of plant diseases have garnered significant attention due to the growing demand for efficient, scalable agricultural solutions. Traditionally, disease detection has relied on visual inspection by agronomists, a method that is time-consuming, labor-intensive, and prone to human error. This has prompted researchers to explore automated approaches using machine learning (ML) and deep learning (DL). Convolutional neural networks (CNNs) have been especially impactful in this domain, achieving remarkable results in image-based plant disease detection. For instance, CNN models have demonstrated over 99% accuracy in classifying various plant diseases [9]. Similarly, Mohanty et al. developed a deep learning model capable of classifying 14 plant diseases with an accuracy of 99.35% [10]. These advancements underscore CNNs' ability to handle large, complex datasets, marking a significant shift toward automated and reliable disease prediction in agriculture [11]. However, recent literature reviews on DL in agriculture have noted gaps, particularly in the application of advanced visualization techniques and modifications to established models to further enhance plant disease identification [12], [13].

Object detection algorithms have further propelled progress in plant disease detection, moving beyond traditional feature extraction methods that depend on color, texture, and shape. These conventional techniques often struggle with general-

ization across diverse datasets and require extensive manual engineering. In contrast, deep learning-based object detection models, such as YOLO (You Only Look Once) and Faster R-CNN, have revolutionized the field by enabling real-time detection with competitive accuracy [14]. YOLO's speed and precision have shown promising results in agricultural settings, where rapid identification of diseases, pests, and anomalies is critical for timely intervention. Faster R-CNN, with its more complex architecture, has been particularly successful in detecting subtle variations in plant health that traditional methods might overlook [15]. These advancements significantly enhance precision agriculture by enabling more accurate and timely responses to emerging plant health issues [16].

The integration of drone technology with deep learning models [17] has opened new opportunities for plant disease prediction at scale. Drones equipped with high-resolution RGB and multispectral cameras provide a unique vantage point, enabling the monitoring of vast agricultural areas. This approach offers higher spatial resolution and more frequent monitoring compared to traditional satellite or ground-based methods. Drones have proven particularly effective for early-stage disease detection, capturing subtle changes in crop health before the disease spreads [18]. While satellite-based monitoring is valuable for large-scale assessments, it is limited by lower spatial resolution and revisit frequency, making it less effective for detecting early-stage diseases [19]. Ground-based methods, though accurate, are labor-intensive and impractical for large-scale operations [20].

Recent studies have demonstrated the effectiveness of com-

binning deep learning models with drone-captured imagery for plant disease detection. CNN-based models have shown high accuracy in identifying plant diseases from drone imagery, even under challenging environmental conditions [21]. This synergy between drone technology and advanced machine learning techniques provides a scalable, efficient approach to crop health monitoring, allowing for timely interventions and optimized resource allocation. Consequently, drones paired with state-of-the-art object detection models are becoming essential tools in precision agriculture, enhancing disease prediction and prevention strategies [11], [22].

III. METHODOLOGY

In this study, we propose a framework for plant disease detection in cashew orchards using drone-acquired imagery and state-of-the-art object detection models. Our approach leverages the CADI-AI dataset, which provides diverse images of cashew plant abnormalities captured under real-world conditions. The methodology is centered around two advanced YOLO models, YOLOv9 and YOLOv10, which are specifically optimized for real-time disease detection. We apply data augmentation techniques to address the challenges of dataset imbalance and environmental variability, while an automated learning process further enhances the model's performance. In the following subsections, we describe the dataset, model architecture used to evaluate the effectiveness of the proposed framework in identifying various stress factors affecting crops, including insect damage, abiotic stress, and diseases.

A. Dataset

For this study, we utilized the "CADI-AI: Cashew Disease Identification" dataset, publicly available on Kaggle. This dataset was specifically developed to address the challenges of detecting various diseases affecting cashew crops using drone technology. All images were captured using a P4 Multi-spectral drone at a resolution of 1600 x 1300 pixels [23]. The dataset comprises 4,736 images, including close-up shots and wider views of cashew plant abnormalities.

The dataset is structured for machine learning experiments, particularly for deep learning models targeting image-based disease detection. It is split into training, validation, and testing sets, each annotated with disease categories such as insect damage, abiotic stress, and pathogenic diseases. These varied categories make the dataset an ideal resource for training object detection models, particularly given the real-world complexities introduced by environmental factors like varying lighting conditions and crop maturity stages [24]. Although comprehensive, further improvements could be made by collecting images from different regions and bloom cycles to enhance the robustness of the models [25].

B. Model Design

In this research, we implemented two object detection models: YOLOv9 and YOLOv10, each designed for real-time detection and optimized for precision agriculture. YOLOv9,

introduced in early 2024, focuses on balancing detection accuracy and speed. Two key innovations distinguish YOLOv9: the Programmable Gradient Information (PGI) framework and the Generalized Efficient Layer Aggregation Network (GELAN).

The PGI framework enhances the backpropagation of gradients during training by facilitating more effective gradient flow through the network, which reduces the vanishing gradient problem commonly encountered in deep networks. For a given layer l , the gradient g_l is computed as:

$$g_l = \frac{\partial L}{\partial W_l} \quad (1)$$

where L is the loss function and W_l are the weights for layer l . PGI enhances this by dynamically scaling the gradients during backpropagation, ensuring that deeper layers receive meaningful updates, thus improving the model's ability to converge on accurate predictions.

The second innovation, GELAN, optimizes the model's architecture by aggregating feature maps from different layers more efficiently. Given feature maps $F_1, F_2, \dots, F_n$, GELAN combines them using an efficient aggregation function:

$$F_{out} = \sum_{i=1}^n \alpha_i F_i \quad (2)$$

where α_i are learnable weights. This technique allows YOLOv9 to reduce its computational footprint while still improving detection accuracy. YOLOv9 achieves a 0.6% increase in Average Precision (AP) on the MS COCO dataset compared to YOLOv8 [26], making it both faster and more accurate for object detection tasks.

YOLOv10 builds on the advancements of YOLOv9, further improving performance in high-resolution imagery and real-time applications. One of its key innovations is the removal of the Non-Maximum Suppression (NMS) dependency. Traditionally, NMS is used to eliminate redundant bounding boxes by keeping the box with the highest confidence score. However, this process can be computationally expensive. YOLOv10 introduces a novel NMS-free training strategy that assigns dual labels to overlapping objects, reducing the reliance on post-processing. The dual-label strategy optimizes the predicted class probabilities p_c and the objectness score p_o such that:

$$p_c^{dual} = \max(p_{c1}, p_{c2}), \quad p_o^{dual} = \max(p_{o1}, p_{o2}) \quad (3)$$

where p_{c1}, p_{c2} and p_{o1}, p_{o2} are the class probabilities and objectness scores for the overlapping objects. This approach effectively balances detection speed and accuracy, significantly enhancing inference times.

YOLOv10 also incorporates architectural improvements, such as lightweight classification heads and spatial-channel decoupled downsampling. The classification head outputs the final predictions using a simplified fully connected layer:

$$C(x) = W_c x + b_c \quad (4)$$

where W_c are the learned weights, x is the input feature vector, and b_c is the bias term. This lightweight architecture reduces computational complexity and enhances adaptability across platforms, from high-powered servers to edge devices.

Finally, YOLOv10 uses a rank-guided block design, where the rank of feature maps is adaptively adjusted to optimize feature representation. This rank-guided adjustment minimizes computational load by focusing on the most salient features while ignoring less relevant information. These architectural upgrades result in lower latency and smaller model size, without compromising detection accuracy, particularly on large datasets like COCO [26].

Both YOLOv9 and YOLOv10 are optimized for real-time performance, making them highly effective in precision agriculture, where rapid and accurate disease detection is essential. The models detect key features such as color patterns, texture variations, and shape distortions, which are critical for identifying stress factors like insect damage, abiotic stress, and pathogenic diseases. Through innovations in gradient optimization, efficient layer aggregation, and the removal of traditional bottlenecks like NMS, these models achieve superior performance in detecting plant diseases from drone-captured imagery.

IV. TRAINING AND EXPERIMENTAL SETUP

The training process for our experiments was conducted using the CADI-AI dataset, consisting of high-resolution drone images of cashew crops. The dataset was split into 70% for training, 15% for validation, and 15% for testing. To enhance model robustness and mitigate overfitting, we employed a variety of data augmentation techniques, including horizontal flipping, vertical flipping, brightness and contrast adjustments, random gamma changes, Gaussian blur, and slight shifts, scaling, and rotations. These augmentations helped simulate diverse environmental conditions, enabling the models to better generalize to real-world scenarios.

We implemented two object detection models, YOLOv9 and YOLOv10, focusing on evaluating their performance under varying environmental factors and dataset imbalance. YOLOv9 was trained for 70 epochs, while YOLOv10 was trained for 100 epochs, both using a batch size of 16 and an input image size of 640x640 pixels. Optimization was performed using the Adam optimizer with an initial learning rate of 0.001, combined with a warm-up strategy. A cosine annealing scheduler was employed to gradually reduce the learning rate during training, facilitating smoother convergence.

The experiments were executed on Google Colab, utilizing an NVIDIA Tesla P100 GPU with 16 GB VRAM to ensure efficient training on the large dataset. The models were implemented using Python 3.10 and deep learning frameworks such as PyTorch and the Ultralytics YOLO library. We also applied mixed precision training to optimize memory usage and computational efficiency, particularly crucial when handling high-resolution drone imagery.

Both models were evaluated using standard metrics, including accuracy, precision, recall, and mean Average Precision

at IoU 0.5 (mAP50), with a particular focus on their ability to detect insect damage, abiotic stress, and plant diseases. Our experiments aimed to investigate whether the architectural improvements in YOLOv9 and YOLOv10 could reduce overfitting and improve generalization compared to previous models, such as YOLOv8 and YOLOv5. The results demonstrate that both models show potential for real-world applications in precision agriculture, particularly in the context of drone-based plant disease detection.

V. RESULTS AND DISCUSSION

This section presents both quantitative and qualitative analyses of the performance of the YOLOv9 and YOLOv10 models on the CADI-AI dataset, highlighting improvements and their impact. Performance metrics, including mean Average Precision (mAP50), precision, and recall, are used to compare these models against previous iterations. Additionally, qualitative visualizations demonstrate the accuracy of the detections.

A. Quantitative Analysis

The performance of YOLOv9 and YOLOv10 was evaluated across three categories: insect damage, abiotic stress, and pathogenic diseases. YOLOv9 achieved an overall mAP50 of 0.601, with its highest performance in detecting insect damage (mAP50 of 0.591). However, it showed lower effectiveness in detecting abiotic stress (0.512) and diseases (0.512). YOLOv10, while slightly behind YOLOv9 in overall mAP50 (0.571), performed better in detecting abiotic damage with an mAP50 of 0.773. Nevertheless, it struggled more with insect stress (0.464) and disease detection (0.476).

Additionally, Table II outlines the experimental procedures and performance of the various models, providing insights into the specific challenges faced, such as overfitting and dataset variability. YOLOv9 displayed better generalization compared to YOLOv10, despite both models encountering overfitting during training.

Figures 2 and 3 present the confusion matrices for YOLOv9 and YOLOv10, respectively. YOLOv9 demonstrated higher precision (0.760) and recall (0.642) than YOLOv10, which had a precision of 0.762 and a recall of 0.617. Both models, however, excelled at detecting insect-related stress, with YOLOv10 achieving a precision of 0.676 and a recall of 0.558, making it particularly effective in identifying insect damage.

B. Qualitative Analysis

To complement the quantitative results, we present qualitative detection outputs. Figures 4 and 5 show sample detections with bounding boxes around stress factors in both training and testing datasets. These visualizations demonstrate the models' ability to accurately detect insect damage, abiotic stress, and disease conditions across different scenarios.

C. Case Studies

Insect damage detection remained the strongest area for both models, with YOLOv10 achieving an mAP50 of 0.713 and YOLOv9 following with 0.591. However, both models

TABLE I
BENCHMARK PERFORMANCE OF ON PLANT DISEASE CLASSES

Model	Insect mAP50	Abiotic Stress mAP50	Disease mAP50	Overall mAP50
YOLOv9	0.591	0.512	0.512	0.601
YOLOv10	0.464	0.773	0.476	0.571

TABLE II
EXPERIMENTAL PROCEDURES USING YOLO ARCHITECTURES

Training Data	Model	Comments	mAP
Original + Background Images	YOLOv8x	Little overfitting	0.60
Original + Background Images	YOLOv8x	More background images; overfitting	0.56
Original Images Only	YOLOv8m	Early overfitting	0.59
Original + Background + Augmented Images	YOLOv8n	converged	0.50
Original + Background + Augmented Images	YOLOv5x	Converged	0.60
Original + Background + Augmented Images	YOLOv5x	Increased batch size improved performance	0.615
Original + Background + Augmented Images	YOLOv5x	Performance improved; converged	0.648
Original Images Only	YOLOv5s	Converged	0.592
Original Images Only +Augmented Images	YOLOv9	Converged	0.601
Original + Augmented Images	YOLOv10	Converged	0.571

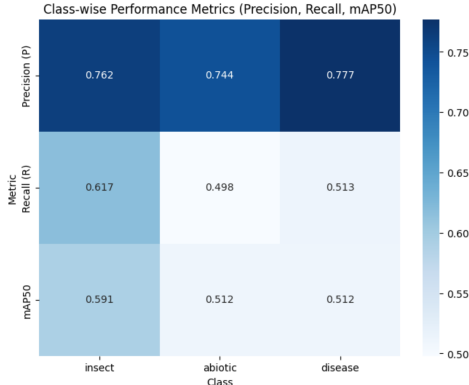


Fig. 2. YOLOv9 Confusion Matrix

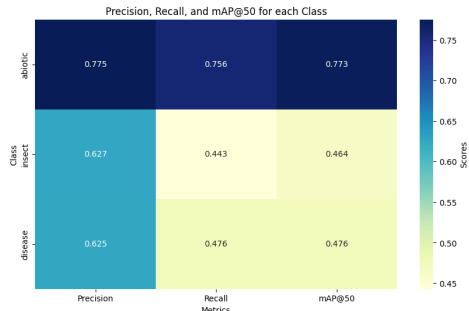


Fig. 3. YOLOv10 Confusion Matrix

struggled more with detecting abiotic stress and diseases, where visual cues were subtler. YOLOv9 scored 0.512 for abiotic stress detection and 0.512 for disease detection, while YOLOv10 scored 0.773 for abiotic stress and 0.476 for disease. These challenges are likely due to the variations in lighting and crop maturity, which obscure visual symptoms



Fig. 4. Sample instances from the training set with bounding boxes around detected stress factors (insect damage, abiotic stress, disease).



Fig. 5. Sample instances from the testing set with bounding boxes around detected stress factors.

and make detection more difficult.

D. Discussion of Findings

Our results indicate that YOLOv10 outperforms YOLOv9 in detecting insect-related stress, while YOLOv9 demonstrates stronger overall performance. The architectural improvements in YOLOv9, such as enhanced gradient flow and feature aggregation, likely contributed to its superior performance in general

object detection. However, both models faced challenges in detecting abiotic stress and diseases, which are characterized by more subtle visual cues. Additionally, variations in lighting and environmental conditions further affected the models' ability to consistently detect these stress factors.

Future research should focus on improving the models' ability to generalize across diverse environmental conditions, such as varying lighting and crop maturity stages. Integrating multispectral or hyperspectral imaging could help detect more subtle stress factors, such as abiotic stress, which are often invisible in standard RGB images. Additionally, deploying these models for real-time analysis during drone flights could facilitate more timely interventions, improving overall precision agriculture practices.

VI. CONCLUSION

In this study, we demonstrated the effectiveness of advanced object detection models, specifically YOLOv9 and YOLOv10, using drone-acquired imagery for crop disease detection. By training and evaluating these models, we achieved reasonable accuracy and speed in detecting various types of crop stress, underscoring the potential of drone technology and deep learning in automating agricultural disease detection. However, limitations included a relatively small dataset and environmental variability, such as lighting and weather, which affected model performance. Detection of abiotic stress factors was less accurate due to their subtle visual features. Future work should focus on expanding dataset diversity by incorporating images from a wider range of geographical regions and crop types. Additionally, integrating multispectral or hyperspectral imaging could enhance the models' ability to detect diseases not visible in standard RGB images. Implementing the models in real-time during drone flights could facilitate timely disease detection, allowing for immediate interventions in crop management and improving agricultural outcomes.

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